

REVIEW

Effectiveness of auriculotherapy in reducing stress in healthcare personnel: A systematic review

Efectividad de la auriculoterapia para reducción del estrés en personal de salud: una revisión sistemática

Eficácia da auriculoterapia na redução do estresse em profissionais de saúde: uma revisão sistemática

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Received: 08/02/2025

Accepted: 16/09/2025

Abstract

Introduction: The mental health of healthcare personnel is crucial for providing high-quality patient care. However, the demands and responsibilities of the work environment can cause stress. Auriculotherapy has been shown to be an effective way of reducing stress and promoting the well-being of healthcare personnel. **Objective:** To evaluate the effectiveness of auriculotherapy in reducing stress in healthcare personnel based on the best available evidence. **Methodology:** A systematic review was conducted by searching literature databases such as LILACS, PubMed, the Cochrane Library, SciELO, the Virtual Health Library and Google Scholar. Randomized clinical trials published in Spanish, Portuguese and English between 2012 and 2022 were included, with those with unclear or questionable methodology excluded. Search strategies were developed and ten articles were selected using the Joanna Briggs Institute checklist. Finally, data extraction and synthesis were performed. **Results:** The articles showed that auriculotherapy is an effective way of reducing stress. Of these, 40 % indicated that needle auriculotherapy was more effective; while 20 % suggested that auriculotherapy without a specific protocol was more effective according to the symptoms reported at each consultation. It was also observed in 10% of the studies that the therapist's experience improved the effectiveness of the treatment. **Conclusions:** Auriculotherapy appears to be more effective at reducing stress than the placebo and control groups. Needle auriculotherapy was more effective than seed auriculotherapy.

Key words: Work stress; Auriculotherapy; Acupuncture; Nursing equipment; Healthcare personnel; Occupational health (DeCS).

Resumen

Introducción: La salud mental del personal de salud es fundamental para asegurar una atención de calidad a los pacientes. No obstante, la carga de responsabilidades y exigencias del entorno laboral pueden generar estrés. La auriculoterapia es eficaz para reducir el estrés y favorecer el bienestar del personal de salud. **Objetivo:** Evaluar la efectividad de la auriculoterapia para la reducción del estrés en personal de salud de acuerdo a la mejor evidencia disponible. **Metodología:** Revisión sistemática con búsqueda de literatura en bases de datos como LILACS, PubMed, Cochrane Library, Scielo, Biblioteca Virtual en Salud y buscador Google Académico. Se incluyeron ensayos clínicos aleatorios entre 2012 y 2022 en idiomas español, portugués e inglés, excluyendo aquellos de metodología poco clara y dudosa. Se desarrollaron estrategias de búsqueda, se seleccionaron 10 artículos mediante Lista de Verificación de Joanna Briggs Institute, finalmente se realizó extracción y síntesis de los datos. **Resultados:** Los artículos evidenciaron que la auriculoterapia fue efectiva para la reducción del estrés. De estos, el 40 % señaló que la auriculoterapia con agujas tuvo mayor efectividad, un 20 % de los estudios indicaron que la auriculoterapia sin un protocolo específico mostró mayor efectividad según los síntomas reportados



en cada consulta. En un 10 % de los estudios se observó que la experiencia del terapeuta mejoró la efectividad del tratamiento. **Conclusiones:** La auriculoterapia parece ser más efectiva para reducir el estrés en comparación con los grupos placebo y control. La auriculoterapia con agujas tuvo mayor efectividad que la auriculoterapia con semillas.

Palabras clave: Estrés laboral; Auriculoterapia; Acupuntura; Equipo de enfermería; Personal de salud; Salud ocupacional (DeCS).

Abstrato

Introdução: A saúde mental dos profissionais de saúde é essencial para garantir um atendimento de qualidade ao paciente. No entanto, a carga de responsabilidades e as demandas do ambiente de trabalho podem gerar estresse. A auriculoterapia é eficaz na redução do estresse e na promoção do bem-estar dos profissionais de saúde. **Objetivo:** Avaliar a eficácia da auriculoterapia para redução do estresse em profissionais de saúde de acordo com as melhores evidências disponíveis. **Metodologia:** Foi realizada uma revisão sistemática com busca bibliográfica nas bases de dados LILACS, PubMed, Biblioteca Cochrane, Scielo, Biblioteca Cochrane, Biblioteca Virtual em Saúde e no mecanismo de busca Google Acadêmico. Foram incluídos ensaios clínicos randomizados entre 2012 e 2022 em espanhol, português e inglês, excluindo aqueles com metodologia pouco clara ou duvidosa. Estratégias de busca foram desenvolvidas, 10 artigos foram selecionados usando a lista de verificação do Joanna Briggs Institute e a extração e síntese de dados foram realizadas. **Resultados:** Os artigos demonstraram que a auriculoterapia foi eficaz na redução do estresse. Destes, 40 % indicaram que a auriculoterapia com agulhas foi mais eficaz. 20 % dos estudos indicaram que a auriculoterapia sem um protocolo específico foi mais eficaz, com base nos sintomas relatados em cada consulta. 10 % dos estudos observaram que a experiência do terapeuta melhorou a eficácia do tratamento. **Conclusões:** A auriculoterapia parece ser mais eficaz na redução do estresse em comparação aos grupos placebo e controle. A auriculoterapia com agulhas foi mais eficaz do que a auriculoterapia com sementes.

Palavras-chave: Estresse no trabalho; Auriculoterapia; Acupuntura; Equipe de enfermagem; Pessoal de saúde; Saúde ocupacional (DeCS).

Introduction

One of the main responsibilities of healthcare personnel is to provide quality care and attention. However, this responsibility can be highly stressful ⁽¹⁾. The stress affecting the mental health of healthcare workers is a significant and worrying problem that brings with it cultural, economic and social challenges ⁽²⁾, and has a negative effect on quality, efficiency and productivity at work. This



results in a decrease in patient satisfaction. It also causes chronic disorders that affect not only the body and mind, but also personal and family life and education ⁽²⁾.

Globally, healthcare workers experience stress levels that are up to 25.8 % higher than those observed in the general population ⁽³⁾. In Latin American countries, 60 % of healthcare workers suffer from stress related to their working conditions ⁽⁴⁾. They face various challenges, including interprofessional relationships, managing patient suffering and pain, lack of supplies and equipment, high demand, long working hours, violence, low wages and precarious conditions ^(5,6). In addition, a lack of personal growth opportunities, poor leadership from superiors, high exposure to infectious diseases or substances are risk factors that can trigger high levels of stress ⁽⁷⁾. Stress is the body's natural and innate response to threats and challenging stimuli, and can manifest to varying degrees ⁽⁸⁾. It can generate physiological and psychological responses that alter the body's normal state ⁽⁹⁾.

The presence of stressful stimuli activates a survival mechanism that causes the release of hormones such as epinephrine and norepinephrine. These hormones increase blood pressure, heart rate and blood glucose levels ⁽¹⁰⁾. Furthermore, repeated exposure can have a negative impact on health ⁽¹¹⁾. This can manifest as body pain, lack of energy, changes in eating habits, sleep problems, heart disease, anxiety, migraines, addictions and obesity ^(12,13). In this context, integrative and complementary health practices (ICHP) are a set of therapeutic approaches, services and methods oriented towards a holistic approach ⁽¹⁴⁾. They promote measures designed to reduce stress levels and enhance overall well-being ⁽⁵⁾.

Auriculotherapy is an ICHP originating from traditional Chinese medicine (TCM) ⁽⁶⁾ that involves stimulating specific points on the ear ⁽¹⁵⁾. These points correspond to different parts of the body and stimulating them can produce physiological responses such as the release of neurotransmitters that improve mood and relieve pain, as well as participating in other bodily functions. It can also



lead to the release of cytokines that optimize immune system function, reduce inflammation in the body, and activate neurological reflexes that promote balance ⁽¹⁶⁾.

Several studies have demonstrated the effectiveness of auriculotherapy in treating stress ^(16,17) and in the care of healthcare personnel ^(5,18-20). It is a simple, inexpensive treatment that is easy to apply and has few or no side effects ⁽¹⁵⁾.

No updated systematic review of clinical trials specifically studying auriculotherapy and stress variables in healthcare personnel was found in the literature review conducted. Addressing this knowledge gap is crucial, as leading research into practices such as auriculotherapy will strengthen nursing's contribution to advancing knowledge in ICHP in healthcare. This will consolidate nursing as a discipline focused on the individual, not just the disease. Adopting this technique would not only significantly reduce stress among healthcare personnel, but also positively impact the quality of care they provide to patients.

In light of the above and the need to address stress through natural prevention and recovery mechanisms with a comprehensive approach, the objective was set to evaluate the effectiveness of auriculotherapy for reducing stress in healthcare personnel, based on the best available evidence.

Methodology

This systematic review was conducted in accordance with the steps outlined in the Cochrane Manual: a) Formulating the research question and developing the inclusion criteria. b) Searching for relevant studies. c) Selecting studies and extracting data. d) Assessing the risk of bias in the included studies. e) Analyzing the data. f) Assessing reporting bias. g) Presenting the results. h) Interpreting the results and conclusions.

First, the population, intervention, outcomes (PIO) components were used to formulate the research question. The elements of the question were identified as follows: P = healthcare personnel, I = auriculotherapy and O = stress reduction. Therefore, the systematic review question was: How



effective is auriculotherapy in reducing stress in healthcare personnel, based on the best available evidence?

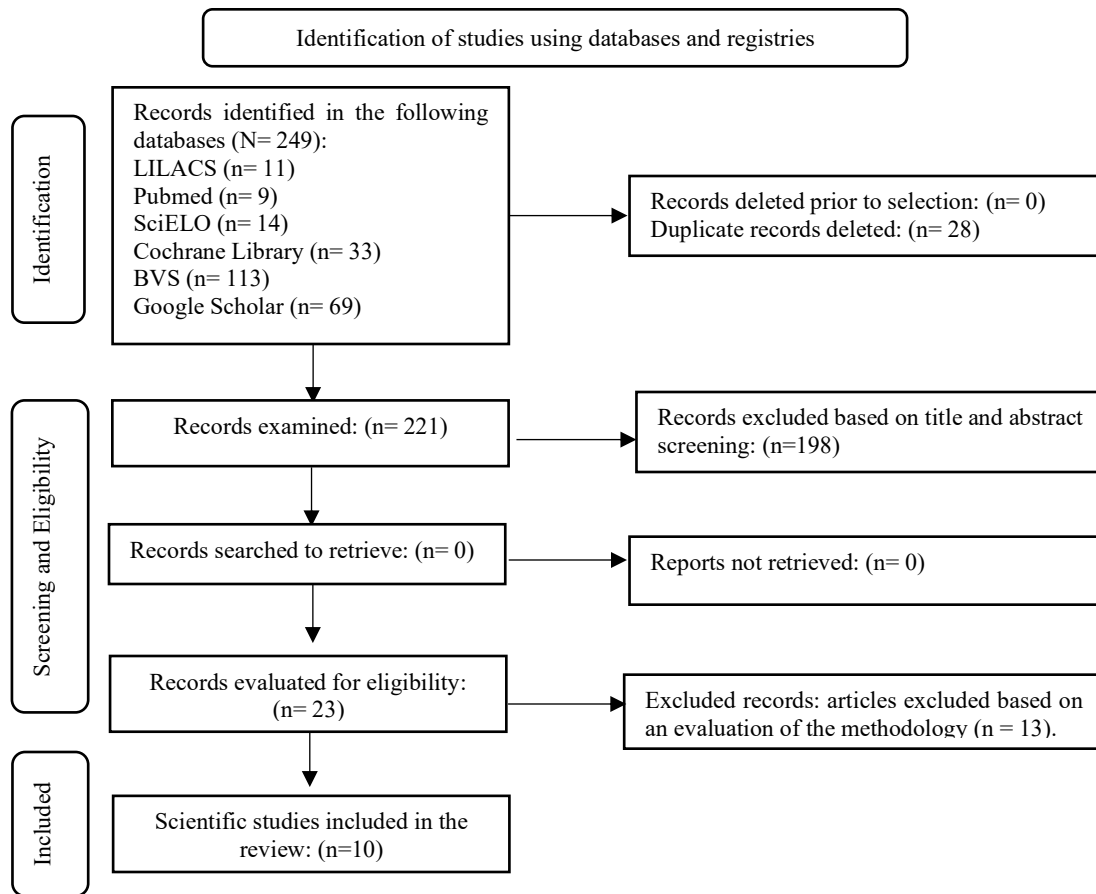
We included Randomized Clinical Trials (RCTs) published in indexed journals in Spanish, Portuguese, and English between 2012 and 2022, as well as studies that included auriculotherapy and stress variables in healthcare personnel. Articles with unclear or dubious methodology, grey literature studies (reports, working papers, theses and government documents) and duplicate studies across the different databases were excluded.

The search strategy used the LILACS, PubMed, Cochrane Library, SciELO, Virtual Health Library (VHL) and Google Scholar databases. The Boolean operators AND and OR were employed alongside the descriptors identified in Medical Subject Headings (Mesh) and Health Sciences Descriptors (DeCS). The combinations used were: Auriculotherapy AND stress AND nursing AND clinical trial, acupressure OR auriculotherapy AND stress AND nursing AND health personnel, and auriculotherapy OR ear acupuncture OR auricular acupuncture AND psychological stress OR stress.

A total of 249 articles were found, and those that did not meet the selection criteria were eliminated. A critical reading and evaluation of the studies was then carried out during the analysis of the information using the Joanna Briggs Institute critical appraisal checklist for RCT⁽²¹⁾. Subsequently, ten studies were selected, and a descriptive synthesis of their results was carried out to allow a detailed description of their characteristics (author, year of publication, country, objectives, hypotheses, sample, results obtained, participant dropout rate and effectiveness of the interventions). Finally, the PRISMA 2020 flow diagram was included, (Figure 1).



Figure 1. PRISMA flow diagram, 2024



Source: Self-developed

Results

In terms of the characteristics of the studies, it was found that 90 % were conducted in Brazil ^(5,6,18,22–27), while the remaining 10 % were conducted in South Korea ⁽²⁰⁾. Of the studies that examined the population 90 % included nursing professionals ^(5,6,18,20,22–27), while the remaining 10 % included nursing students ⁽²³⁾. Another study ⁽²⁴⁾ included nursing assistants and technicians in addition to nursing professionals. A further study ⁽⁶⁾ covered a more diverse population, including nursing technicians, dentists, dental assistants, community health workers and physicians, (Table 1).



Table 1. Characteristics of the included studies, 2024 (n=10)

Author and Year	Country	Journal	Population and Sample Size
Kurebayashi, et al. ⁽⁵⁾	Brazil	Rev. Esc. Enferm USP.	Nursing professionals: (n= 75).
Kurebayashi, et al. ⁽²²⁾	Brazil	Acta Paul Enferm.	Nursing professionals: (n= 49).
Prado, et al. ⁽²³⁾	Brazil	Rev. Latino-Am. Enfermagem.	Nursing Students: (n= 71).
Kurebayashi, et al. ⁽²⁴⁾	Brazil	Rev. Latino-Am. Enfermagem.	Nursing professionals, nursing assistants, and nursing technicians: (n= 75).
Kurebayashi, et al. ⁽²⁵⁾	Brazil	Rev. Eletr. Enfermero.	Nursing professionals: (n= 75).
Kurebayashi, Da Silva ⁽¹⁸⁾	Brazil	Rev. Latino-Am. Enfermagem.	Nursing professionals: (n= 175).
Kurebayashi, et al. ⁽²⁶⁾	Basil	Rev. Bras Enferm.	Nursing professionals: (n= 175).
Prado, et al. ⁽²⁷⁾	Brazil	Rev. Esc. Enferm USP.	Male nurses: (n= 168).
Lee, et al. ⁽²⁰⁾	South Korea	Complementary Therapies in Clinical Practice.	Outpatient nurses: (n= 54).
Damasceno, et al. ⁽⁶⁾	Brazil	Rev. Latino-Am. Enfermagem.	Nurses, nursing technicians, dentists, dental assistants, community health workers, and physicians: (n= 49).

Source: Self-developed

Table 2 details the characteristics of the interventions, including the treatments applied to different auriculotherapy points, the materials used, stimulation times, the number and duration of sessions, and follow-up after the interventions. The main results of the different studies revealed that auriculotherapy was more effective with needles than with seeds ^(5,24,25). Auriculotherapy administered by experienced therapists was found to effectively reduce stress ⁽²²⁾. Two studies ^(23,27) showed that auriculotherapy was more effective than placebo, contrary to the findings of one study ⁽²⁰⁾, which, despite showing a reduction in stress, found no differences between the experimental group and the placebo group. On the other hand, auriculotherapy without a protocol, i.e., individualized, was more effective than auriculotherapy with a protocol for stress reduction ⁽¹⁸⁾. However, another study obtained the opposite result, reporting a reduction in stress with both auriculotherapy with and without a protocol.



Table 2. Characteristics of the interventions, 2024 (n=10)

Author/Year	Objective	Interventions		Main Outcomes
		Experimental Group (EG)	Control Group (CG)	
Kurebayashi, et al. ⁽⁵⁾	Assess stress levels in hospital nursing staff and analyze the effectiveness of auriculotherapy with needles and seeds.	G2: needles (n=27). G3: seeds (n=27).	G1: Without any intervention (n=22).	The EG showed statistical differences ($p < 0.05$) from the second assessment onwards. Auriculotherapy was found to reduce stress in nursing professionals, with better results achieved using needles than seeds. No statistically significant results were found in the CG.
Kurebayashi, et al. ⁽²²⁾	Evaluate the effectiveness of auriculotherapy using semi-permanent needles in reducing stress levels among hospital nursing staff.	G2: semi-permanent needles with less experienced therapists (n= 16). G3: semi-permanent needles with experienced therapists (n= 17).	G1: Without any intervention (n=16).	A statistically significant difference was found between G1 and G3 ($p=0.036$), demonstrating that stress in nursing professionals was effectively reduced by auriculotherapy administered by more experienced therapists.
Prado, et al. ⁽²³⁾	Evaluate whether real auriculotherapy or placebo auriculotherapy is more effective at reducing stress levels.	G1: auriculotherapy (n= 24).	G2: Without auriculotherapy (n=25) G3: Placebo group with simulated points (n=22).	The G1 group had a Cohen's effect size of 1.81 (considered to be a high effect) and demonstrated a statistically significant difference after eight sessions, which was still evident at the follow-up stage ($p < 0.001$). This shows that true auriculotherapy is more effective at reducing stress.
Kurebayashi, et al. ⁽²⁴⁾	Evaluate the effectiveness of auriculotherapy in reducing stress, and analyze how coping mechanisms change during the initial and final stages of treatment.	G2: needles (n=27). G3: seeds (n=26).	G1: Control (n= 22).	There were no statistically significant differences in stress levels between the G3 group with seeds and the other groups. However, there were differences within the G3 group over time ($p < 0.05$). Auriculotherapy with needles decreased stress levels ($p < 0.05$). At the fourth assessment, differences were observed between the control group and the G2 group with needles in the distancing domain ($p < 0.05$).
Kurebayashi, et al. ⁽²⁵⁾	Identify the traditional Chinese medicine (TCM) diagnoses for the stress symptoms that responded to auriculotherapy.	G2: semi-permanent needles (n= 27). G3: seeds (n= 26).	G1: control without treatment (n=22).	Statistical differences were found after treatment ($p= 0.023$) between group G2 (needles) and the control group, showing that auriculotherapy reduced stress with better results using needles than seeds.



Kurebayashi, Da Silva ⁽¹⁸⁾	Evaluate the efficacy of auriculotherapy, both with and without a protocol, in lowering stress levels among nursing staff.	G2: auriculotherapy with protocol (specific established points) using semi-permanent needles (n= 58). G3: auriculotherapy without protocol (chosen based on the symptoms reported at each session) using semi-permanent needles (n= 59).	G1: Control (n= 58)	Statistical differences were found between CG and EG in both evaluations ($p < 0.05$). G2 with protocol achieved an index of 0.79 (large effect) with an average reduction in stress levels (27 %). G3 without protocol achieved a large reduction in stress (37 %), equivalent to a Cohen's index of 1.21 (very large effect). It was demonstrated that auriculotherapy without protocol (individualized) reduced stress more effectively in nursing staff.
Kurebayashi, et al. ⁽²⁶⁾	Evaluate the effectiveness of auriculotherapy in improving quality of life and reducing stress among nursing staff.	G2: auriculotherapy with protocol (specific established points) using semi-permanent needles (n= 58) G3: auriculotherapy without protocol (chosen based on the symptoms reported at each session) using semi-permanent needles (n= 59).	G1: Control (n= 58)	Statistical differences were observed in the list of stress symptoms in the second assessment between the CG and the two EG ($p = 0.000$), and in the third assessment, between the CG and the group with protocol ($p = 0.004$) and between the CG and the group without protocol ($p = 0.002$). Both the EG with protocol and the EG without protocol reduced stress ($p < 0.05$).
Prado, et al. ⁽²⁷⁾	Verify the effectiveness of experimental auriculotherapy and placebo auriculotherapy with simulated points in the treatment of stress in nurses at a charitable hospital in São Paulo.	G1: auriculotherapy.	G2: Placebo simulated points G3: Control without intervention.	Experimental auriculotherapy achieved a 43 % reduction in stress, demonstrating statistical significance after eight sessions ($p < 0.001$). Placebo auriculotherapy achieved a 26 % reduction in stress, with a statistically significant difference observed after 12 sessions ($p < 0.001$). The CG without intervention did not achieve any reduction in stress.
Lee, et al. ⁽²⁰⁾	Investigate the impact of auricular acupressure on stress, anxiety and depression experienced by outpatient nurses.	G1: with seeds (n=28).	G2: Placebo with seeds (n=26).	Although no reduction in stress was observed, a decrease in cortisol was recorded. However, no statistically significant differences were found between the EG and the CG.

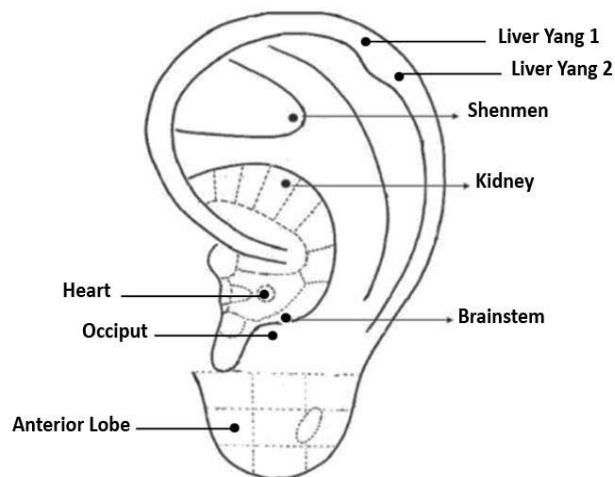


Damasceno, et al. ⁽⁶⁾	Evaluate the effectiveness of auriculotherapy in alleviating work-related stress experienced by Family Health Strategy workers during the Coronavirus (Covid-19) pandemic.	G1: auriculotherapy (n=25).	G2: Placebo group (n=24).	The auriculotherapy group showed reductions in work-related stress.
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Source: Self-developed

In terms of auriculotherapy, 100 % of studies in the intervention groups stimulated the Shenmen point ^(5,6,15,20,22–27), 90 % stimulated the brainstem point ^(5,6,18,22–27), 60 % stimulated the kidney point ^(5,18,22,24–26), 20 % stimulated liver yang points 1 and 2 ^(18,26), and 10 % stimulated the heart, occipital, and anterior lobe points ⁽²⁰⁾. Notably, one study ⁽²⁷⁾ used the Shenmen point in all participants, reaffirming its importance in reducing stress, (Figure 2).

Figure 2. Auriculotherapy points used in the different studies



Source: Adapted from Kurebayashi, et al. 2012 ⁽⁵⁾

Meanwhile, 30 % of the studies in the control group involved stimulating points on the wrist ^(6,20,23) and the outer ear ^(6,23,27). Ten per cent of the studies involved stimulating points on the face ⁽²⁷⁾, hip, elbow and shoulder ⁽²⁰⁾. Twenty per cent of the studies involved stimulating points based on symptoms reported by participants in each session according to TCM diagnoses ^(18,26). These

studies attributed the effectiveness of auriculotherapy to the stimulation of auricular points. One study explained that the ear has rich innervation and that its stimulation activates neurological mechanisms mediated by the parasympathetic nervous system, which promotes the release of neurotransmitters in the body ⁽²³⁾. Similarly, another study ⁽⁵⁾ detailed that the stimulated points activate myelinated nerve fibers that send impulses to the spine, brain, pituitary gland, and hypothalamus, causing the release of endorphins into the bloodstream.

Of the studies investigating auriculotherapy, 70 % used semi-permanent needles ^(5,18,22-26). 30 % used mustard seeds ^(5,6,24). 20 % used vaccaria seeds ^(20,27) and 10 % used unspecified seeds ⁽²⁵⁾. Knowing which materials are used enables us to determine the most effective method of reducing stress.

Regarding the duration of the intervention sessions, it was observed that 40 % of the studies involved a total of eight sessions, one per week, each lasting between five and ten minutes ^(5,22,24,25). Similarly, 12 sessions were conducted twice a week, each lasting between 5 and 10 minutes ^(18,23,24,27). In contrast, one study ⁽²⁰⁾ implemented sessions once a week for five weeks in both groups, but no reduction in stress levels was observed. The researchers emphasized the need for follow-up over a period of more than eight weeks to obtain meaningful results. Another study ⁽⁶⁾ mentioned using between three and six sessions.

The additional instructions provided in the various studies had a direct impact on the effectiveness of auriculotherapy during the intervention. It is therefore worth mentioning that, in 20 % of cases, participants were instructed to stimulate the seeds three times a day with moderate pressure, and to remove the needles either 24 hours before the consultation, or if they experienced any discomfort, such as itching or an allergic reaction ^(5,24). A further 20 % mentioned using a spot locator ^(5,23). 10 % said that they pressed the seeds continuously for five days, for five seconds at a time with a two-second pause every five minutes, four times a day (at 07:00, 12:00, 17:00 and 22:00). They were



then instructed to remove the points on the sixth day and gently massage their ears. Additionally, 10 % used hypoallergenic tape ⁽¹⁸⁾, microporous tape ⁽²⁶⁾ and hypoallergenic microporous adhesive tape ⁽⁶⁾. Some researchers emphasized the importance of hand washing with soap and water, (Table 4).

Table 4. Intervention protocol, 2024 (n = 10)

Author/Year	Auriculotherapy Points	Devices/Materials	Duration of the Sessions	Additional Instructions
Kurebayashi, et al. ⁽⁵⁾	G2 and Shenmen, kidney, and brainstem.	G3 Semi-permanent needles and mustard seeds.	G2 and G3: 8 sessions, one per week, each lasting between 5 and 10 minutes.	Stimulate the seeds three times a day with moderate pressure and remove the needles 24 hours before the consultation, or if discomfort such as itching or allergy occurs.
Kurebayashi, et al. ⁽²²⁾	G2 and Shenmen, kidney, and brainstem.	G3 Semi-permanent needles.	G2 and G3 underwent 8 sessions, one per week, each lasting between 5 and 10 minutes.	A point locator was used.
Prado, et al. ⁽²³⁾	G1: Shenmen and brainstem. G3: Wrist and outer ear.	Semi-permanent needles.	G2 and G3: 12 sessions, one per week, each lasting between 5 and 10 minutes.	A point locator was used.
Kurebayashi, et al. ⁽²⁴⁾	G2 and Shenmen, kidney, and brainstem.	G3 Semi-permanent needles and mustard seeds.	G2 and G3: 8 sessions, one per week, each lasting between 5 and 10 minutes.	Stimulate the seeds three times a day with moderate pressure and remove the needles 24 hours before the consultation, or if discomfort such as itching or allergy occurs.
Kurebayashi, et al. ⁽²⁵⁾	G2 and Shenmen, kidney, and brainstem.	G3 Sterilized needles and seeds.	G2 and G3: 8 sessions, one per week, each lasting between 5 and 10 minutes.	Hand washing with soap and water.
Kurebayashi, Silva ⁽¹⁸⁾	Da G2: Shenmen, brain stem, kidney, liver yang 1 and 2. G3: The points were chosen based on the symptoms reported in each consultation, according to TCM diagnoses.	Semi-permanent needles.	G2 and G3: 12 sessions, twice a week, each lasting between 5 and 10 minutes.	The needles were applied with hypoallergenic tape.



Kurebayashi, et al. ⁽²⁶⁾	G2: Shenmen, brain stem, kidney, liver yang 1 and 2. G3: Points were selected based on the patient's response to treatment.	Semi-permanent needles.	G2 and G3: 12 sessions, twice a week, each lasting between 5 and 10 minutes.	The needles were applied with micropore tape.
Prado et al. ⁽²⁷⁾	G1: Shenmen and brainstem. G2: External area of the ear and face.	Vaccaria seeds.	12 sessions twice a week, 15-day follow-up.	
Lee, et al. ⁽²⁰⁾	G1: Shenmen, heart, occiput, anterior lobe. G2: Wrist, hip, elbow, shoulder.	Vaccaria seeds.	Both groups attended one session per week for five weeks. In addition, the application was alternated between each ear weekly.	The seeds were pressed continuously for five days, five seconds at a time, with a two-second pause every five minutes, four times a day (at 7 a.m., 12 p.m., 5 p.m., and 10 p.m.). On the sixth day, the points were removed and the ears were gently massaged.
Damasceno, et al. ⁽⁶⁾	G1: Shenmen and brainstem point. G2: Wrist and outer ear.	Mustard seeds.	3, 6 sessions.	Hypoallergenic microporous adhesive tape was used.

Source: Self-developed, Note: G1= Group 1, G2= Group 2, G3= Group 3, DX= Diagnosis, TCM= Traditional Chinese Medicine.

With regard to adverse effects, 60 % of the studies reported no such effects in participants ^(6,18,20,24,26,27). One study ⁽²³⁾ reported that a participant withdrew due to hypersensitivity and pain in the ear area. Meanwhile, 20 % of the studies reported that a participant who had recently undergone bariatric surgery withdrew because auriculotherapy caused food cravings and restlessness ^(5,25). A further 20 % reported that a participant withdrew due to experiencing nightmares following auriculotherapy sessions. It should be noted that this participant worked nights and received treatment at the end of their shift ^(22,25). One study ⁽²⁴⁾ mentions that some participants reported discomfort due to the placement and permanence of the needles; however, no major adverse effects occurred except for itching, which disappeared when the needles were replaced with seeds.



Discussion

This systematic review aimed to evaluate the effectiveness of auriculotherapy in reducing stress in healthcare personnel, based on the best available evidence. The studies indicated that healthcare personnel may experience stress as a result of their working conditions. This finding coincides with that reported by the World Health Organization ⁽²⁸⁾, which states that high levels of stress affect healthcare and care workers in response to staff shortages, low wages, unsafe and inadequate working conditions, extremely stressful environments and a lack of job security.

In terms of theoretical reference, auriculotherapy is based on the fundamental principles of traditional Chinese medicine. This approach maintains that the ear is connected to twelve meridians, or energy channels through which Qi, or vital energy, circulates. It is believed that stimulating the ear helps to restore the balance between Qi and blood ^(6,15). Lazarus and Folkman (1984) explain in their transactional model that, based on the theoretical foundation of stress, stress depends not only on the situation itself, but also on the person's subjective evaluation of it. Stress arises when a person continuously assesses environmental demands and perceives them as exceeding their coping capacity, generating psychological, physiological and behavioral responses ⁽²⁹⁾. In this sense, stimulating the ear points could explain its effectiveness in reducing stress, when stimulated, reflex connections are generated with other parts of the body through neural pathways. This results in viscerosomatic reflexes that act to restore homeostasis to the hypothalamic–pituitary–adrenal axis, which is responsible for the release of the cortisol hormone ⁽³⁰⁾.

At the same time, stimulation of the points activates small myelinated nerve fibers, which transmit impulses to the spine, brain, pituitary gland and hypothalamus. This causes endorphins to be released into the bloodstream, generating a feeling of relief and well-being ⁽²²⁾. Therefore, studies on stress and auriculotherapy indicate that this technique is beneficial for reducing stress. This is



supported by the results of a meta-analysis ⁽³⁰⁾, which states that auriculotherapy reduces stress levels and anxiety in healthcare personnel.

The most commonly used auriculotherapy points were Shenmen, the brainstem point and the kidney point. These points were shown to significantly reduce stress. These findings coincide with those reported by researchers ⁽³⁰⁾, who, despite noting a lack of standardization in auriculotherapy use, recommended the use of points such as Shenmen, the brainstem, the kidney, the sympathetic nerve, the lung and the liver. In addition to reducing stress, these points can be used to treat anxiety and burnout syndrome in healthcare professionals. This is because the Shenmen point has been shown to have a calming effect, promoting relaxation and sedation ^(31,32). Meanwhile, the Kidney point has an energetic function, while the Brain Stem point has calming properties ⁽¹⁸⁾.

Regarding the materials or devices used in auriculotherapy, studies suggest that semi-permanent needles are more effective than seeds. Using semi-permanent needles means the points remain active continuously without the individual needing to stimulate them. This result is consistent with the findings of other researchers ⁽³⁰⁾, who reported that semi-permanent needles were more effective than other materials or groups in reducing the evaluated outcomes, since they were found to be effective in reducing stress when compared with control or placebo groups. This may be because, compared to seeds, semi-permanent needles provide more consistent and deeper stimulation.

Using seeds requires participants to follow a series of additional instructions, which can lead to variability in results ⁽³⁰⁾. This is because if participants forget to stimulate the seeds at the indicated frequency and for the indicated duration, the technique's effectiveness could decrease. Additionally, the amount of pressure applied to the seeds by participants will vary. This demonstrates that needles require fewer additional instructions than seeds. When applied, needles cause pain and local inflammation, keeping the point active without the need for manual stimulation ⁽³³⁾.



Effective intervention sessions typically lasted between 5 and 10 minutes, with one to two sessions taking place each week for a period ranging from 8 to 12 weeks. This is consistent with the findings of one study ⁽³⁰⁾, in which the researchers recommended eight sessions, one per week. This reflects the lack of a standardized protocol for sessions, stimulation points and instructions that have already been tested in clinical research.

This study only presents evidence of the efficacy of auriculotherapy across different studies, without making comparisons between techniques or protocols. This is due to the heterogeneity in the selection of points, devices and materials used, duration of sessions, additional instructions and adverse effects recorded during interventions, which can be attributed to the technique being applied in different countries. The variable results of the studies limited the possibility of performing a meta-analysis. Furthermore, as this review was based on the PIO framework rather than the PICO approach, no comparisons were made between the different techniques.

The lack of in-depth studies in Mexico analyzing the acceptance of the intervention and its combination with other therapies makes it difficult to evaluate the most effective intervention for reducing stress among healthcare personnel. It is recommended that nursing staff develop and standardize a protocol that has been tested and validated in the Mexican population, taking hygiene and biosafety aspects into account ⁽³⁰⁾. It should be noted that nursing staff play a key role in primary care and are responsible for contributing to the development of innovative healthcare strategies.

Conclusions

This review shows that auriculotherapy is an effective way of reducing stress in healthcare personnel. However, the limited number of studies and the absence of updated systematic reviews made it difficult to evaluate and generalize the results. Nevertheless, based on these findings, it is possible to replicate the study to strengthen research and the nursing body of knowledge, as it is



essential to incorporate alternatives such as auriculotherapy, which are minimally invasive and have few side effects.

Therefore, healthcare professionals, especially nursing staff, must reach a unanimous consensus to promote research that encourages the implementation of ICHP as an alternative to conventional methods in the care and treatment of patients and healthcare personnel. These practices must strengthen the body's natural recovery mechanisms, both physical and mental.

Ultimately, multicenter studies are recommended to compare auriculotherapy with other ICHP, evaluating its long-term effects and developing new areas of nursing research.

Conflict of interest

The authors stated that they have no conflict of interest.

Funding

The main researcher (CVU number 1109514) would like to express their gratitude for the scholarship awarded by the Ministry of Science, Humanities, Technology and Innovation (Secretaría de Ciencia, Humanidades, Tecnología e Innovación - SECIHTI) to pursue doctoral studies in Nursing Sciences within the Mental Health and Risk Behaviours in Vulnerable Populations academic body at the Facultad de Enfermería Mochis.

Artificial intelligence

The authors stated that they have not used any artificial intelligence resources in any of the sections of this manuscript.

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How to reference this article: Valenzuela-Urrea G, Ahumada-Cortez J, Gámez-Medina M, Apodaca-Armenta R, Gómez-Rodríguez G. Effectiveness of auriculotherapy in reducing stress in healthcare personnel: A systematic review. SANUS [Internet]. 2025 [cited dd mmm yyyy];10(21):e561 Available at: DOI/URL

